real-world objects — it enhances the immersive experience for end users through its 3D rendering.

When USD grows in the metaverse world as a standard facility, Forrester predicts that growth and standardisation of the four technologies listed below can elevate the growth of the metaverse.

Extended reality (XR) is a combination of VR, AR and mixed reality (MR). The continual convergence of these technologies will enhance the immersive user experience.

Web 3.0 will enable delivery of next-generation web technology solutions. NFTs and secured blockchain based network solutions will help create resilient, high performing and secure platforms for metaverse based cryptocurrency transactions.

Zero Trust Edge (ZTE) will provide seamless and uninterrupted network facilities for metaverse solutions. It will combine high performance in network communications with zero or low latency and have embedded security features.

TuringBot is a term coined by Forrester for automatic and autonomous code development and testing automation. It can solve problems and develop solutions based on in-built AI-powered software.

Token economy and metaverse solutions

We have been hearing about NFTs and cryptocurrencies in recent times, but they are misconceptualised as an investment option and are interchangeable. There is also a thin line drawn between Web 3.0, token economy and digital currencies in association with NFTs and cryptocurrencies. It is better to understand each of them to understand how and when to use them.

Token economy refers to a digital transformation process using blockchain technology, where physical assets are digitised and processed in a blockchain platform for online trading. In simple terms, blockchain platform enables token economy models using

Figure 1: Technology convergence for metaverse solution

cryptocurrency tokens. Smart contracts are used to programmatically define how tokens can be managed, exchanged and realised in a blockchain platform.

Industrial metaverse

According to a report from Market Prospects, the industrial metaverse is expected to touch US$ 540 billion in revenue by 2025.

The industrial metaverse prepares current industrial infrastructure for digital transformation. It gives a central platform for companies, offering security, easy accessibility and usability.

Industrial metaverse integrates all processes and applications, bringing everything under a single roof, and making it transparent and manageable.

Companies can fully utilise and leverage the potential of industrial metaverse in the following ways:

1. Brand awareness via factory walks and virtual interactions
2. Virtual presentations of all sorts of products and services
3. Payment via blockchain/NFTs or metaverse's crypto
4. R&D collaborations using digital twins for design, simulation and testing of new products
5. Strong feedback services using customer's voice

The industrial metaverse is still at an infant stage, and it faces challenges because of high equipment costs and technical difficulties.

The idea of an industrial metaverse has attracted a great deal of interest over the last 12 months. It describes, in short, a highly immersive and connected virtual and physical reality enabling never-before-seen levels of intelligent connectivity and AI. END

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